EXPERIMENT 15:

Perform transformation using Direct Linear Transformation.

AIM:- Transformation using Direct Linear Transformation.

PROGRAM:-

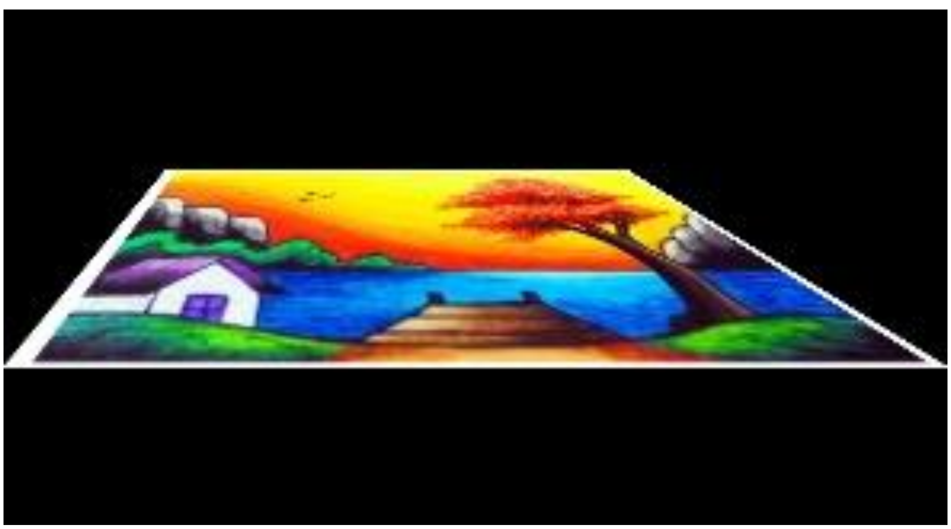
```
import cv2
import numpy as np
import os
image_path = r'C:\Users\pavan\OneDrive\Opencv\nature.png'
if not os.path.exists(image_path):
    print("Image file not found at:", image_path)
    exit()
img = cv2.imread(image_path)
if img is None:
    print("Failed to load image. Check file format or path.")
    exit()
rows, cols = img.shape[:2]
src_points = np.float32([
    [0, 0],
    [cols - 1, 0],
    [0, rows - 1],
    [cols - 1, rows - 1]
])
dst_points = np.float32([
    [50, 50],
    [cols - 100, 50],
    [0, rows - 50],
    [cols - 1, rows - 50]
])
H, mask = cv2.findHomography(src_points, dst_points, method=0)

if H is not None:
```

```
    transformed_img = cv2.warpPerspective(img, H, (cols, rows))
    cv2.imwrite('DLT_transformed.jpg', transformed_img)
    print("DLT Transformation complete. Output saved as DLT_transformed.jpg")
else:
    print("Homography could not be computed.")
```

INPUT:-

Output:-



RESULT:-

Thus, the program is executed successfully